THESE ARE SOME EXAMPLES OF TYPICAL QUESTIONS – THEY WILL NOT BE INCLUDED IN THE MIDTERM – DO NOT MEMORIZE

MIDTERM HEADER:

Midterm; Duration: 1 hour 20 minutes
(CLOSED BOOK – NO CALCULATORS)

Name: _____ Total: _____/ 80 marks

Student No:_____

Interpretation of the questions is part of your mark. Please **do not ask questions**. If you think that something is unclear or ambiguous, **make a reasonable assumption** (one that does not contradict the question), write it at the start of your answer, and solve the problem.

In order to help you with time management, the points for each question are allocated based on expected time needed to solve the question (i.e., if a question is 5 points, it should take you around 5 minutes to complete; 80 minutes = 80 points). Manage your time carefully.

SAMPLE QUESTIONS

If an input pin connected to an active high source of interrupt, these should be configured as: Question 1 options:

- ○ Falling edge interrupt
- ◉ Rising edge interrupt
- ○ Level interrupt

To test whether an active low button (P1.4) is pressed, we would code: Question 2 options:

- if(~(P1->IN & (uint8_t)(1 << 4)))
- if(!(P1->IN & (uint8_t)(1 << 4)))
- if((P1->IN & (uint8_t)(1 << 4))) if((P1
- ->IN & (uint8_t)(0 << 1)))

Complete the missing code (underlined)

```
PORT1_IRQHandler() {
    if(P1->IFG & (uint8_t)(1 << 2)) {
        P1->IFG &=~(uint8_t)(1<< 2);
     if(P2->OUT & (uint8_t)(1 << 1))
          P2->OUT &= ~(uint8_t)(1 << 6);
      else
           P2->OUT |= (uint8_t)(1 << 6);
    }
}
```

ARM Processors have 16 registers called R0 through R15. Match each register set with the purpose of each register set Question 12 options:

1. R13
2. R0-R12
3. R15
4. R14

Register that keeps the address of the instruction to be executed	3
Register that keeps the address of the current stack	1
Register that keeps the return address when handling subroutine calls	4
General Purpose Registers	2

The C code volatile char a = 2 & 1; will make a equals to

0

The C code volatile char b = 2 && 1; will make b equals to

1

The C code volatile char c = 2 | 7; will make c equals to

7

The C code volatile char d = 0; d |= 1<<2; will make d equals to

4

The C code volatile char e = 0; e |= 128>>1; will make e equals to

64

Write a few lines of C code that decrement the bottom 4 bits of P1OUT, without afecting any other bits. For example, if P1OUT were 0x25 before the operation, its new value should be 0x24. If P1OUT were 0x60 before the operation, its new value should be 0x6F (notice in this example the bottom 4 bits went into underflow, i.e., 0-1=15, without affecting the top 4 bits).

(I give documentation for the microcontroller; you read and answer these questions)

i. How much memory does it have, and of what kind? How many channels for I/O and what kind? What are the main features of the CPU? How many registers? What size are they?

ii. Give a list of specialized Input/Output devices found on the microcontroller. Which ones of those you would not find normally in a PC?

iii. What is the size of the Data Bus? What does this mean?

Explain the concept of *masking* interrupts. Discuss different masking mechanisms and how they can affect the execution of a program.

Describe difficulties you might encounter when developing an embedded real-time applications using Java as the development programming language. Also, describe advantages of using Java for such applications.

Sometimes embedded systems need real-time performance. Explain how a RISC processor like the ARM can help in guaranteeing such performance predictably (hint: think about how RISC processors execute instructions). (DIFFICULT QUESTION – FOR THOSE AIMING TO GET A+)

You are designing the system for controlling the traffic lights at an intersection. Give one reason why you would use a microcontroller rather than a microprocessor for this system.

What is a non-maskable interrupt? How it is detected? What is the difference with a maskable interrupt?

Explain the steps of the interrupt service cycle (the idea is for you to explain the important concepts of this hardware mechanism)